

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC Engineering Services

Subject: Electronics & Telecommunication Engineering

Questions: 150

Source: Official source PDF extraction

AI-generated PYQs: 0

1. [UNKNOWN] unknown

1. Consider a common-emitter current gain of $\beta = 150$ and a base current of $I_B = 15 \mu A$. If the transistor is biased in the forward active mode, the collector and emitter current will be 2.25 mA and 2.27 mA 3.25 mA and 2.27 mA 2.25 mA and 1.37 mA 3.25 mA and 1.37 mA

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2. [UNKNOWN] unknown

1. 3 Q 2.3 52 3.6 Q A temperature-sensing device can be modeled as a first-order system with a time constant of 6 s. It is suddenly subjected to a step input of $25^\circ C$ - $150^\circ C$. The indicated temperature in 10 s after the process has started will be

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3. [UNKNOWN] unknown

1. 3 kHz and 54 5% (c) Logical instructions 1. 6 kHz and 46.5% (d) Machine control instructions 1.3 kHz and 46.5%/c. 4

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4. [UNKNOWN] unknown

1. A pseudoinstruction is a machine instruction. $D = e^{-x} \sin y$ $a_e^{-x} \cos y$ $ay + 2z$ a, C/M2 ? A pseudoinstruction is an instruction to the assembler. The ORG (origin) is an example of pseudoinstruction. It is not possible to use ORG more than once in a program. Select the correct answer using the code given below. 1 and 3 2 and 3 1 and 4 2 and 4 8 nC 4 nC 2 nC 1 nC

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5. [UNKNOWN] unknown

1. It is similar to GSM. appends a 150-bit header, followed by the network layer which uses a 120-bit

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6. [UNKNOWN] unknown

2. The input to a bridge rectifier is 230 V (r.m.s.), 50 Hz. The d.c. output voltage and the ripple factor with R_L of 100 Ω and capacitor filter of 1000 μF are 207 V and 0.028 325 V and 0.028 207 V and 0.020 325 V and 0.020

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7. [UNKNOWN] unknown

2. 5 V (13) 3.4 V A A 4.3 V 0.008 5.2V The magnitude of A_o for the system to be oscillatory will be (a) 0.2

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8. [UNKNOWN] unknown

2. It allows each station to transmit header. Network layer packets are over the entire frequency spectrum transmitted via two networks, each of all the time. which uses a 26-bit header. The

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9. [UNKNOWN] unknown

2. Needless transmission power and streams are multiplexed to form 1 T2 have to deal with local interference stream and 7 T1 streams are only multiplexed to form 1 T3 stream.

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10. [UNKNOWN] unknown

2. 7×10^{-5} 3.2×10^{-5} 4.9×10^{-5} physical layer, network layer and application layer monitoring data-link layer, presentation layer and session layer monitoring transport layer, session layer and application layer monitoring

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11. [UNKNOWN] unknown

3. The effect of reduction in effective base width due to increase in reverse voltage of BJT is Hall effect Early effect Zener effect Miller effect

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12. [UNKNOWN] unknown

3. It assumes that multiple signals destination network only accepts up to add linearly. 900 bits long. The number of bits, including headers delivered to the Select the correct answer using the code (a) 2706 bits (la) 2634 bits 2554 bits 2476 bits 1 and 2 only 1 and 3 only (a) 2 and 3 only (d) 1, 2 and 3

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13. [UNKNOWN] unknown

3. Frequency planning and infrastruc- Further 6 T3 streams are multiplexed ture needed to form 1 T4 stream. If each Ti stream is of 1- 544 Mbps, the data rate of 1 T4.

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14. [UNKNOWN] unknown

4. What is the drain current for a D-MOSFET having the characteristic values IDss of 10 mA, Vasmo of -4 V and IIDs of $+2$ V? 22.5 mA 17.5 mA 12.5 mA 2.5 mA non-inverting op-amp feedback op-amp high-gain op-amp

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15. [UNKNOWN] unknown

4. These require both circuit stream should be switching and packet switching (a) 211. 8 Mbps Select the correct answer using the code given below. (b) 232.6 Mbps 1, 2 and 4 1, 3 and 4 243.4 Mbps (c) 1, 2 and 3 (d) 274 .2 Mbps (d) 2, 3 and 4 destination network, is given below. VBY-S-ELX/51A 25 [P.T.O. ===== PAGE 26 =====

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16. [UNKNOWN] unknown

5. In the Wien bridge oscillator, the 00 phase-shift is met by using lead-lag network and by using (a) inverting op-amp

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17. [UNKNOWN] unknown

6. What is the frequency of oscillation for an RC phase-shift oscillator with R of 10 k Ω and C of 0.001 μ F in each of its three RC sections? 5.0 kHz 5.5 kHz 6.0 kHz (d) 5.5 kHz

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18. [UNKNOWN] unknown

6. 3 A 5 -4 A VBY-S-ELX/ 51A 3 [P.T.O. ===== PAGE 4 =====

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19. [UNKNOWN] unknown

7. When there is no clock signal applied to CMOS logic circuits, they are referred to as complex CMOS logic circuits static CMOS logic circuits NMOS transmission gates random PMOS logic circuits

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20. [UNKNOWN] unknown

8. One form of NMOS circuit logic that minimizes power dissipation and maximizes device density is called pass transistor logic sequential logic circuit NMOS SRAM cell NMOS transmission gate VBY—S—ELX/51A 2 ===== PAGE 3 =====

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21. [UNKNOWN] unknown

10. A d.c. voltage supply provides 60 V when the output is unloaded. When connected to a load, the output drops to 56 V. The value of the voltage regulation is 3.7% 5.7% 7.1% 9.1%

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22. [UNKNOWN] unknown

11. In optical communication, the maximum angle in which external light rays may strike the air/glass interface and still propagate down the fiber is called as critical angle numerical aperture angle of refraction acceptance angle 243 mW/m² 232 mW/m² 221 mW/m² 210 mW/m²

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23. [UNKNOWN] unknown

12. The light intensity 3 m from a lamp that emits 25 W of light energy will be 1 infinite voltage gain and zero input impedance infinite voltage gain and infinite bandwidth zero voltage gain and infinite CMRR zero output impedance and zero CMRR

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24. [UNKNOWN] unknown

13. Two resistances, one of 30 Ω and another of unknown value, are connected in parallel. The total power dissipated in the circuit is 450 W when the applied voltage is 90 V. The unknown resistance is 45 Ω 35 Ω 30 Ω 20 Ω

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25. [UNKNOWN] unknown

14. An electric kettle contains 1.5 kg of water at 15 °C. It takes 15 minutes to raise the temperature of water to 95 °C. If the heat loss due to radiations and heating the kettle is 15 kcalories and the supply voltage is 100 V, the current taken will be

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26. [UNKNOWN] unknown

15. A heater element is made of nichrome wire having resistivity equal to $100 \times 10^{-8} \Omega \text{ m}$ and diameter of 0.4 mm. The length of the wire required to get a resistance of 40 Ω will be nearly 9 m 7 m 5 m 3 m

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27. [UNKNOWN] unknown

16. A car is travelling at 72 km/h. If the length of an axle is 2 m and the vertical component of the earth's magnetic field is 40 ab/m², the e.m.f. generated in the axle of the car is 26 mV 2.2 mV

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28. [UNKNOWN] unknown

17. In a telephone receiver, the size of each of the two poles is 1.2 cm x 0.2 cm and the flux between each pole and the diaphragm is $3 \times 10^{-8} \text{ Wb}$. The force attracted to the poles will be nearly 0.15 N 0.20 N

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29. [UNKNOWN] unknown

18. An inductor of 0.5 H inductance and 90 Ω resistance is connected in parallel with a 20 μF capacitor. A voltage of 230 V at 50 Hz is maintained across the circuit. The total power taken from the source is nearly 588 W 145 W 135 W 125 W

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30. [UNKNOWN] unknown

19. A 240 V shunt motor with the armature resistance of 0.1 Ω runs at 850 r.p.m. for an armature current of 70 A. If its speed is to be reduced to 650 r.p.m., the resistance to be placed in series for an armature current of 50 A is nearly 0.82 Ω 1.14 Ω 1.24 Ω

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31. [UNKNOWN] unknown

20. A 200 V d.c. shunt motor with armature resistance of 0.2 Ω and carrying a current of 50 A is running at 960 r.p.m. If the flux is reduced by 10% at constant torque and with negligible iron and friction losses, the speed will be nearly 1280 r.p.m. 1170 r.p.m. 1100 r.p.m. 1060 r.p.m. VBY-S-ELX/51A 4 ===== PAGE 5 =====

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32. [UNKNOWN] unknown

21. Which of the following statements are

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33. [UNKNOWN] unknown

22. Consider the following materials : Lead peroxide Sponge lead Dilute sulphuric acid Gassing occurs at both electrodes. The terminal voltage is 2.6 V. The specific gravity of the electrolyte is 1.21. Select the correct answer using the code given below. 1 and 2 only 1 and 3 only 2 and 3 only 1, 2 and 3

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34. [UNKNOWN] unknown

23. Which of the following statements are correct for d.c. shunt motor? correct for a fully charged lead-acid cell? Speed of a shunt motor is sufficiently constant. For the same current input, its starting torque is not as high as that of a series motor. The motor can be directly coupled to a load such as a fan whose torque increases with speed. Select the correct answer using the code given below. 2 and 3 only 1 and 3 only 1 and 2 only 1, 2 and 3

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35. [UNKNOWN] unknown

24. Which of the following statements are correct for synchronous motors? Synchronous motors are well-suited for direct connection to reciprocating compressors. Over-excited synchronous motors are most commonly used for power factor improvement. Synchronous motors are generally used for current regulation of long transmission lines. Which of the above are active materials of a lead-acid battery? 1 and 2 only 1 and 3 only 2 and 3 only 1, 2 and 3 Select the correct answer using the code given below. 1, 2 and 3 1 and 3 only 1 and 2 only 2 and 3 only VBY-S-ELX/51A 5 [P.T.O. ===== PAGE 6 =====

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36. [UNKNOWN] unknown

25. Which crystal system requires six lattice

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37. [UNKNOWN] unknown

26. The minimum cation-to-anion radius ratio for the coordination number 3 is 0.175 0.155 0.135 0.115

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38. [UNKNOWN] unknown

27. Which of the following materials are categories of ceramic materials? Oxides—Alumina, Zirconia Non-oxides—Carbides, Borides, Nitrides and Silicides Composites—Particulate reinforced combinations of oxides and

non-oxides Select the correct answer using the code given below. 1 and 2 only 1 and 3 only 2 and 3 only 1, 2 and 3
 3.2×10^{25} 3.2×10^{25} 2.2×10^{25} 2.2×10^{25}

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39. [UNKNOWN] unknown

28. Consider the following data for copper : parameters to fully specify its unit cell? Energy for vacancy formation Triclinic is 0.9 eV/atom Monoclinic Atomic weight is 63.5 g/mol Cubic Density is 8.4 g/cm^3 at 1000°C The equilibrium number of vacancies Hexagonal per cubic meter at 1000°C will be

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40. [UNKNOWN] unknown

29. Which of the following are electrical insulating materials? Lucite Mica Bakelite Select the correct answer using the code given below. 1 and 2 only 1 and 3 only 2 and 3 only 1, 2 and 3

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41. [UNKNOWN] unknown

30. The magnitude of the energy gap for an insulator is less than 1 eV between 2 eV to 3 eV more than 3 eV between 1 eV to 2 eV VBY—S—ELX/51A 6 ===== PAGE 7 =====

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42. [UNKNOWN] unknown

31. In a 440 V, 50 Hz transformer, the total iron loss is 3700 W. When the applied voltage is 220 V at 25 Hz, the total iron loss is 750 W. The eddy current loss at the normal voltage and frequency will be (a) 1000 W 1200 W 1400 W 1850 W

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43. [UNKNOWN] unknown

32. A transformer core is wound with a coil carrying an alternating current at a frequency of 50 Hz. The hysteresis loop has an area of 60000 units, when the axes are drawn in units of $10^{-4} \text{ Wb m}^{-2}$ and 10^2 A m^{-1} . If the magnetization is uniform throughout the core volume of 0.01 m^3 , then the hysteresis loss will be 200 W 230 W 270 W 300W

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRONICS-TELECOMMUNICATION-ENGINEERING-ESEP-19_0.pdf

44. [UNKNOWN] unknown

33. The process of evaporating a metal in an inert atmosphere and allowing it to condense on the surface of a cold finger, which is kept at liquid nitrogen temperature of 77 K, is known as d.c. arc method gas-phase condensation sonohydrolysis flame pyrolysis

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45. [UNKNOWN] unknown

34. Which one of the following materials is having the highest electrical conductivity at room temperature? Silver Copper Gold Platinum

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46. [UNKNOWN] unknown

35. Consider the following processes : Sol-gel process Electrodeposition Plasma-enhanced vapour decomposition Gas-phase condensation Sputtering technique The above processes are related to analysis of nano-powders sintering of nano-powders synthesis of nano-powders microwave sintering of nano- powders

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47. [UNKNOWN] unknown

36. In the superconducting state, the flux lines of a magnetic field are ejected out of the superconductor as per Curie effect Faraday's effect Maxwell's effect Meissner effect

48. [UNKNOWN] unknown

37. A null type of instrument as compared to a deflection type of instrument has a higher accuracy is less sensitive is more rugged is faster in response VBY-S-ELX/51A 7 1 P.T.O. ===== PAGE 8 ===== 10 μ 52 20 ptgl 301152 40 452 118.2 °C 126.4 °C 134 • 6 °C 142 8 °C

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49. [UNKNOWN] unknown

38. A Wheatstone bridge requires a change of 7 52 in the unknown resistance arm of the bridge to produce a change in deflection of 3 mm of the galvanometer. The sensitivity and the deflection factor will be nearly 0.23 mm/52 and 2•3 52/mm 0.43 mm/52 and 2•3 52/mm 0.23 mm/52 and 1•3 52/mm 0.43 mm/Q and 1.3 Q/mm

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50. [UNKNOWN] unknown

39. The galvanometer used in a Wheatstone bridge as a detector can detect a current as low as 0.1 nA and its resistance is negligible compared to internal resistance of the bridge. Each arm of the bridge has a resistance of 1 k.Q. The input voltage applied to the bridge is 20 V. The smallest change in the resistance that can be detected is A resistance is determined by voltmeter- ammeter method. The voltmeter reads 100 V with a probable error of ± 12 V and the ammeter reads 10 A with a probable error of ± 2 A. The probable error in the computed value of the resistance will be nearly 0•5 52

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51. [UNKNOWN] unknown

40. The inductance of a 25 A electrodynamic ammeter changes uniformly at the rate of 0 003511H/degree. The spring constant is 10-6 N m/degree. The angle of deflection at full scale will be (a) 135° 04 125° 115° 105°

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52. [UNKNOWN] unknown

43. In a parallel circuit having two branches, the current in one branch is 4 = 10 $\pm$ 2 A and in the other is 1.2 -= 200 $\pm$ 5 A. Considering errors in both /3. and 12 as limiting errors, the total current will be 300 $\pm$ 5 A 300 $\pm$ 6 A 300 $\pm$ 7 A 300 $\pm$ 8 A VBY-S-ELX/51A 8 ===== PAGE 9 =====

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53. [UNKNOWN] unknown

44. A 0-150 V voltmeter has a guaranteed accuracy of 1% of full-scale reading. The voltage measured by this instrument is 75 V. The limiting error will be 5% 4% 3% 2%

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54. [UNKNOWN] unknown

45. A quartz piezoelectric crystal having a thickness of 2 mm and voltage sensitivity of 0.055 V m/N is subjected to a pressure of 1. 5 MN/m2. The voltage output will be 165 V 174V 183 V 192 V

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55. [UNKNOWN] unknown

46. A resistance wire strain gauge with a gauge factor of 2 is bonded to a steel structural member subjected to a stress of 100 MN/m2. The modulus of elasticity of steel is 200 GN/m2. The change in the value of gauge resistance due to the applied stress will be 0•05% 0.10% 030% 0•60%

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56. [UNKNOWN] unknown

47. The applications of photomultipliers are seen in night vision equipment, medical equipment mechanical counters,

timers translational, optical instruments ultrasonic transducer, infrared imaging

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57. [UNKNOWN] unknown

48. A capacitance of 250 pF produces resonance with a coil at a frequency of 106 Hz, while at the second harmonic of this frequency, resonance is produced by a capacitance of 50 pF. The self-capacitance of the coil will be nearly 16.7 pF
20.5 pF 24.3 pF 28.1 pF

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58. [UNKNOWN] unknown

49. Consider the following data for twigs and links : N = Number of nodes L = Total number of links B = Total number of branches
The total number of links associated with a tree is $B - N + 1$ $B - N - 1$ $B + N + 1$ $2B - N + 1$ VBY-S-ELX/51A 9 [P.T.O. ===== PAGE 10 ===== reverse voltage ra impedance to and transfer reverse current ra do and transfer

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59. [UNKNOWN] unknown

50. In ABCD parameters, A and Care called

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60. [UNKNOWN] unknown

51. A coil having resistance of 10 Ω and inductance of 1 H is switched on to a direct voltage of 100 V. The steady-state value of the current will be 10 A 15 A 20A 25 A

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61. [UNKNOWN] unknown

52. A coil has $R = 10 \Omega$ and $L = 15 \text{ FL}$ The voltage at the instant when the current is 10 A and increasing at the rate of 5 A/s will be 5v 2H (a) 125 V The current $i(t)$ after switching will be (a) $5e^{-5t}$ (b) $150 V 4e^{-5t}$ 175 V (c) $3e^{-5t}$ (d) 200 V (d) e^{-5t} VBY-S-ELX/51A 10 ===== PAGE 11 =====

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62. [UNKNOWN] unknown

53. Consider the following RC circuit. The capacitor has an initial charge q_0 such that it opposes the flow of charging current : (a) reverse current ratio and transfer admittance impedance (d) reverse voltage ra admittance do and transfer
The response of the circuit $i(t)$ will be (a) $(E - ffil) e^{-1 \pm c R PC}$

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63. [UNKNOWN] unknown

54. In the following circuit, switch K is thrown from position A to position B at time $t = 0$, the current having previously reached its steady state : A

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64. [UNKNOWN] unknown

55. What is the condition for reciprocity and symmetry in Y-parameter representation? (a) $Y_{21} = Y_{11}$ and $Y_{22} = X'_{21}$

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65. [UNKNOWN] unknown

56. In hybrid parameters, h_{11} and h_{21} are called as input impedance and forward current gain reverse voltage gain and output admittance input impedance and reverse voltage gain output impedance and forward current gain

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66. [UNKNOWN] unknown

57. Consider the following equations : $6V_2 - 4I_2 = 7V_2 - 2I_2$ A, B, C and D parameters are

67. [UNKNOWN] unknown

58. A supply of 250 V, 50 Hz is applied to a series RC circuit. If the power absorbed by the resistor be 400 W at 160 V, the value of the capacitor C will be nearly (a) 30- 5 [IF (b) $Y_{21} = Y_{12}$ and $Y_{11} = Y_{22}$ 41.5 pF 1721 = Y_{22} and Y_u Y_{22} (c) 64 -0 1.1F (d) 1711 Y_{22} and $Y_{21} = '1'_{22}$ (d) 76.8 IS

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68. [UNKNOWN] unknown

59. A 50 Hz sinusoidal voltage $V = 311 \sin \omega t$ is applied to an RL series circuit. If the magnitude of resistance is 5 Ω and that of the inductance is 0.02 H, the r.m.s. value of the steady-state current and the relative phase angle are nearly 19.6 A and 51.5° 27.4 A and - 51.5° 19.6 A and - 51.5° 27.4 A and 51.5°

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69. [UNKNOWN] unknown

60. In a series RC circuit, the values of $R = 10 \Omega$ and $C = 25 \text{ nF}$. A sinusoidal voltage of 50 MHz is applied and the maximum voltage across the capacitance is 2.5 V. The maximum voltage across the series combination will be nearly (a) 6, -4 Ω , 7 mho and -2 (a) 172.7V (b) 6, 4 Ω , 7 mho and 2 184•5V -6, 4 0, -7 mho and 2 (c) 196.3 V (d) 6, 4 Ω , -7 mho and -2 (d) 208•1V VBY-S-ELX/ 51A 11 [P.T.O. ===== PAGE 12 =====

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70. [UNKNOWN] unknown

61. The peak-to-peak ripple voltage for a half-wave rectifier and filter circuit operating at 60 Hz, which has a 680 μF reservoir capacitor, an average output of 28 V and 200 Ω load resistance, will be nearly

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71. [UNKNOWN] unknown

62. The components of full-wave voltage doubler circuit are (b) 0.1 (c) 5 , 0 (a) 2 diodes and 1 capacitor (d) 10.0 (b) 4 diodes and 1 capacitor

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72. [UNKNOWN] unknown

63. An amplifier has a signal input voltage V_i of 0.25 V and draws 1 mA from the (c) 103V source. If the amplifier delivers 8 V to a load of 10 mA, the power gain is (d) 8•1V (a) 340

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73. [UNKNOWN] unknown

64. Three amplifiers of gain $A = (NZ - 60^\circ 2$ are connected in tandem. The feedback loop is closed through a positive gain of 0.008 . (a)

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74. [UNKNOWN] unknown

65. The output voltage from a 5-bit ladder type DAC that has a digital input of (c) 2 diodes and 2 capacitors 11010, and by assuming 0 = 0 V and 1 = + 10 V, is nearly (d) 4 diodes and 2 capacitors (a)

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75. [UNKNOWN] unknown

66. An 8-bit D/A converter has step size of 20 mV. The full-scale output and the resolution will be nearly (b) 320 51V and 0.3% 4 • 6 V and 0-4% 250 (c) 5.1 V and 0-4% (d) 150 (d) 46 V and 0,3% VBY-S-ELX/51A 12 ===== PAGE 13 =====

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76. [UNKNOWN] unknown

68. Consider the following expression : $A \cdot B \cdot C \cdot D + A \cdot B \cdot \bar{C} + A \cdot B \cdot C \cdot \bar{D} + A \cdot \bar{B} \cdot C \cdot D + A \cdot B \cdot C \cdot D + A \cdot \bar{B} \cdot \bar{C} \cdot D + A \cdot B \cdot \bar{C} \cdot \bar{D} + A \cdot \bar{B} \cdot C \cdot \bar{D} + A \cdot \bar{B} \cdot \bar{C} \cdot \bar{D}$
The simplification of this by using theorems of Boolean algebra will be $A + B + C + D$

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77. [UNKNOWN] unknown

69. An electric power generating station supplies power to three loads A, B and C. Only a single generator is required when any one load is switched on. When more than one load is on, an auxiliary generator must be started. The Boolean equation for the control of switching of the auxiliary generator will be $A + B + C$

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78. [UNKNOWN] unknown

70. Which one of the following types of C = 0.01 μ F, $R_A = 10k\Omega$, $R_B = 50k\Omega$, instructions will be used to copy from the frequency and the duty cycle will be the source to the destination location? nearly (a) Arithmetic instructions (a) 1.6 kHz and 54.5% (b) Data transfer instructions

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79. [UNKNOWN] unknown

71. A cascaded arrangement of flip-flops, where the output of one flip-flop drives the clock input of the following flip-flop, is known as synchronous counter ripple counter ring counter up counter

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80. [UNKNOWN] unknown

72. The number of flip-flops required to construct an 8-bit shift register will be 32 16 8 4

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81. [UNKNOWN] unknown

73. Which one of the following specifications does not fit for a single-mode fiber? The bandwidth is 1 GHz/km. The digital communication rate is excess of 2000 Mbytes/s. More than 100000 voice channels are available. The mode field diameter (MFD; spot size) is larger than the core diameter. VBY-S-ELX/ 51A 13 [P.T.O. ===== PAGE 14 ===== 1 3 1 2 and 4 entropy is 1 bit/symbol 2 1 bit/symbol 1-1 bits/symbol 4 1 1-2 bits/symbol 1 P(C) = . 4

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82. [UNKNOWN] unknown

74. For a binary FSK signal with a mark frequency of 49 kHz, a space frequency of 51 kHz and an input bit rate of 2 kbps, the peak frequency deviation will be

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83. [UNKNOWN] unknown

75. A random process $X(t)$ is defined as $X(t) = 2 \cos(2\pi f t + Y)$ where Y is a discrete random variable with $P(Y = 0) = -1/2$ and $P(Y = \pi) = 1/2$. The mean $\mu(1)$ is (a)

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84. [UNKNOWN] unknown

76. A source produces three symbols A, B and C with probabilities $P(A) = 1/2$. The source

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85. [UNKNOWN] unknown

77. An AM wave with modulation index 0.8 has total sideband power of 4.85 kW. The carrier power and the total power radiated will be nearly 0.5 kW 1.0 kW 2.0 kW 4.0 kW 12.2 kW 15.2 kW 12.2 kW 15.2 kW and 20 kW and 20 kW and 25 kW and 25 kW

86. [UNKNOWN] unknown

78. A 360 W carrier is simultaneously modulated by two audio waves with modulation percentages of 55 and 65 respectively. The effective modulation index and the total power radiated are 0.85 and 490.5W 0.65 and 490.5 W 0.85 and 450.5W 0.65 and 450.5W

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87. [UNKNOWN] unknown

79. An amplitude modulated amplifier has a radio frequency output of 50 W at 100% modulation. The internal loss in the modulator is 10 W. The unmodulated carrier power is 40 W 50W 60 W 80W 1 4 VBY-S-ELX/51A 14 ===== PAGE 15 =====

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88. [UNKNOWN] unknown

81. For Gaussian and White channel noise, the capacity of a low-pass channel with a usable bandwidth of 3000 Hz and $B = 10^3$ at the channel output will be 15000 bits/s 20000 bits/s 25000 bits/s 30000 bits/s

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89. [UNKNOWN] unknown

82. For a PM modulator with a deviation sensitivity $K = 2.5$ rad/V and a modulating signal $v_m(t) = 2 \cos(2\pi 2000t)$, the peak phase deviation m will be 1.25 rad 2.5 rad (e) 5.0 rad (d) 7.5 rad the difference signal is larger in amplitude than actual signal more quantization levels are needed the successive samples of signal often differ in amplitude

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90. [UNKNOWN] unknown

83. In a PCM system, non-uniform signal-to-noise ratio of 29 dB, a noise quantization leads to figure of 4 dB and an FM improvement (a) increased quantizer noise factor of 16 dB, the pre-detection and post-detection signal-to-noise ratios are (b) simplification of the quantization process 25 dB and 41 dB (c) higher average SNR 30 dB and 49 dB (d) increased bandwidth 25 dB and 49 dB

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91. [UNKNOWN] unknown

84. The bandwidth required in DPCM is less than that of PCM because 30 dB and 41 dB (a) the number of bits per code is reduced resulting in a reduced bit rate •

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92. [UNKNOWN] unknown

85. For the given transfer function $Y(s) = \frac{1}{s^2 + 3s + 2}$ the response $y(t)$ for a step input $r(t) = u(t)$ will be $\frac{1}{2}e^{-t} - \frac{1}{2}e^{-2t}$ where $u(t)$ is a unit step input. (a) VBY-S-ELX/51A 15 [P.T.O. ===== PAGE 16 =====

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93. [UNKNOWN] unknown

86. The price for improvement in sensitivity

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94. [UNKNOWN] unknown

87. Consider a feedback system with the characteristic equation $1 + K(s+1)(s+2)$ The asymptotes of the three branches of root locus plot of this system will form the following angles with the real axis 60° , 120° and 300° 60° , 120° and 180° 60° , 180° and 300° 40° , 120° and 200° 0 and 180° and 0 and 180°

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95. [UNKNOWN] unknown

88. If the characteristic equation of a feedback control system is given by $s^4 + 20s^3 + 15s^2 + 2s + K = 0$ then the range of values of K for the system to be stable will be $1 < K < 2.49$ $0 < K < 1.49$ $1 < K < 4.49$ $0 < K < 3.49$ 1 function is s^3 . Which of the above statements are correct? 1 and 2 only 1 and 3 only 2 and 3 only 1, 2 and 3 VBY-S-ELX/51A 16 ===== PAGE 17 =====

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96. [UNKNOWN] unknown

89. For a Type-2 system, the steady-state by the use of feedback is paid in terms errors for unit step and unit ramp input of are loss of system gain rise of system gain improvement in transient response, delayed response poor transient response

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97. [UNKNOWN] unknown

90. Consider the following statements regarding a parabolic function : A parabolic function is one degree faster than the ramp function. A unit parabolic function is defined as $.1q$, for $t > 0$ 0 otherwise Laplace transform of unit parabolic

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98. [UNKNOWN] unknown

91. Consider the following open-loop transfer function : $K(s+2) G - (s+1)(s+4)$ The characteristic equation of the unity negative feedback will be $(s+1)(s+4)+K(s+2)=0$ $(s+2)(s+1)+K(s+4)=0$ $(s+1)(s-2)+K(s+4)=0$ $(s+2)(s+4)+K(s+1)=0$

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99. [UNKNOWN] unknown

92. The magnitude and phase relationship between the sinusoidal input and the steady-state output of a system is called as magnitude response transient response steady-state response frequency response

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100. [UNKNOWN] unknown

93. A transfer function having all its poles and zeros only in the left-half of the s-plane is called a minimum-phase transfer function a complex transfer function an all-pass transfer function a maximum-phase transfer function Which of the above are correct? 1 and 4 1 and 3 2 and 4 2 and 3 VBY-S-ELX/51A 17 [P.T.O. ===== PAGE 18 ===== The attenuation (magnitude) produced by

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101. [UNKNOWN] unknown

94. The frequency where magnitude M has a peak value in frequency response is known as (a) normalized frequency (6) resonant frequency peak frequency tuned frequency

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102. [UNKNOWN] unknown

95. For a lead compensator having transfer function $(s+z_c) - + G_c(s) = (s+P_c) (s+*) = P_c a= z_c > 1 P_c$

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103. [UNKNOWN] unknown

98. Consider a disk with an average seek time of 4 ms, rotational delay of 2 ms, rotation speed of 15000 r.p.m. and 512-byte sectors with 500 sectors per track. A file occupies all of the sectors on 5 adjacent tracks. After reading the first track, if remaining tracks can be read with no seek lime, then the time required in sequential organization to transfer the file will be nearly 0.01 second 0034 second 034 second 3.4 seconds

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104. [UNKNOWN] unknown

99. Add 8 and 9 in BCD code. a lead compensator at the frequency of maximum phase lead $\phi_{i,m} = -\text{Jai}$; is (a) b - (b)c ."-i- b a (c) VI) - a (d) b Consider the following statements : A computer will have a multiply instruction. Multiply instruction will be implemented by a special multiply unit. 00010111 00010001 01110111 10001001

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105. [UNKNOWN] unknown

100. Convert the binary number 11000110 Which of the following is correct? (a) Both 1 and 2 are not architectural to Gray code. design issues. (a) 00100101 (b) Both 1 and 2 are not organizational issues. (Li) 10100100 (c) 1 is an architectural design issue while 2 is an organizational issue. (c) 11100110 (d) 1 is an organizational issue while 2 is an architectural design issue. (d) 10100101

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106. [UNKNOWN] unknown

101. The decimal value of the signed binary number complement 10101010 expressed in 2's will be -42 -86 -116 -170 VBY-S-ELX/51A 18 ===== PAGE 19 ===== (a) — 11 3

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107. [UNKNOWN] unknown

102. Which of the following statements is/are

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108. [UNKNOWN] unknown

103. In a cache with 64-byte cache lines, how many bits are used to determine which byte within a cache line an address (c) points to? 11 16 2 (d)- 11 8 6

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109. [UNKNOWN] unknown

104. A system has 64-bit virtual addresses and 43-bit physical addresses. If the (a) Assembly language program runs pages are 8 kB in size, the number of faster. bits required for VPN and PPN will be respectively (b) Writing of assembly language programming is easy. Assembly language program is portable. Assembly language program contains less instruction. 51 bits and 30 bits 30 bits and 51 bits 51 bits and 13 bits 30 bits and 13 bits VBY-S-ELX/S1A 19 [P.T.O. ===== PAGE 20 =====

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110. [UNKNOWN] unknown

105. A soft error is a correct? An address generated by the CPU is commonly referred to as a physical address. An address seen by the memory unit is commonly referred to as a logical address. The run-time mapping from virtual to physical address is done by the memory management unit (MMU). Select the correct answer using the code given below. 1 only 2 only 3 only 1, 2 and 3 (a) regular-nondestructive event (Li) random-nondestructive event random-destructive event regular-destructive event

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111. [UNKNOWN] unknown

106. A main memory can hold 3 page frames and initially all of them are vacant. Consider the following stream of page requests : 2, 3, 2, 4, 6, 2, 5, 6, 1, 4, 6 If the stream uses FIFO replacement policy, the hit ratio h will be

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112. [UNKNOWN] unknown

107. Which one of the following is an advantage of assembly language over high-level language?

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113. [UNKNOWN] unknown

108. Which of the following statements are

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114. [UNKNOWN] unknown

109. The vector $\vec{RAB}$ extends from $A(1, 2, 3)$ to B . If the length of $\vec{RAB}$ is 10 units and its direction is given by $\vec{a} = 0.6\vec{a}_x + 0.64\vec{a}_y + 0.48\vec{a}_z$ the coordinates of B will be $7\vec{a}_x + 4.8\vec{a}_y + 4.8\vec{a}_z$, $6\vec{a}_x + 6.1\vec{a}_y + 4.8\vec{a}_z$, $7\vec{a}_x + 7.8\vec{a}_y + 6\vec{a}_z$, $8\vec{a}_x + 8.4\vec{a}_y + 7.8\vec{a}_z$

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115. [UNKNOWN] unknown

110. What is the value for the total charge enclosed in an incremental volume of 10^{-9} m^3 located at the origin if

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116. [UNKNOWN] unknown

111. The unit vector extending from origin toward the point $G(2, -2, -1)$ is $(\frac{2}{3}\vec{a}_x - \frac{2}{3}\vec{a}_y - \frac{1}{3}\vec{a}_z)$

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117. [UNKNOWN] unknown

112. Ground waves progress along the surface of the earth and must be polarized (a) horizontally (b) circularly (c) elliptically (d) vertically

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118. [UNKNOWN] unknown

113. For a lossless line terminated in a short

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119. [UNKNOWN] unknown

114. The characteristic impedance of an 80 cm long lossless transmission line having $L = 0.25 \text{ nH/m}$ and $C = 100 \text{ pF/m}$ will be 25Ω , 40Ω , 52Ω , 80Ω

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120. [UNKNOWN] unknown

115. It is required to match a 200Ω load to a 300Ω transmission line to reduce the SWR along the line to 1. If it is connected directly to the load, the characteristic impedance of the quarter-wave transformer used for this purpose will be 275Ω , 260Ω , 245Ω , 230Ω

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121. [UNKNOWN] unknown

116. For a standard rectangular waveguide circuit, the stationary voltage minima having an aspect ratio of 2:1, the minima and maxima are separated by cutoff wavelength for TE_{10} mode will be nearly (a) $\frac{\lambda_c}{2}$ (b) $\frac{\lambda_c}{4}$

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122. [UNKNOWN] unknown

117. The irises in the rectangular metallic waveguide may be inductive, resistive, capacitive. Select the correct answer using the code given below. 1, 2 and 3; 1 and 2 only; 1 and 3 only; 2 and 3 only

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123. [UNKNOWN] unknown

118. A 10 GHz signal is propagated in a waveguide whose wall separation is 6 cm. The greatest number of half-waves of electric intensity will be possible to establish between the two walls. The guide wavelength for this mode of propagation will be 6.48 cm , 4.54 cm , 2.48 cm , 1.54 cm

124. [UNKNOWN] unknown

119. In TEM, n mode, m and n are integers denoting the number of (a) $\frac{\lambda}{2}$ the wavelengths of intensity between each pair of walls the wavelengths of intensity $\frac{\lambda}{3}$ between each pair of walls $\frac{\lambda}{4}$ the wavelengths of intensity between each pair of walls $\frac{\lambda}{8}$ the wavelengths of intensity between each pair of walls

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125. [UNKNOWN] unknown

120. Consider the following statements with reference to dipole arrays : In broadside array, all the dipoles are fed in the same phase from the same source. In end-fire array, the magnitude of the current in each element is same and there is no phase difference between these currents. Which of the above statements is/are correct? 1 only 2 only Both 1 and 2 Neither 1 nor 2

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126. [UNKNOWN] unknown

121. Which of the following are the advantages of Silicon over Insulator (SOI)? Lower diffusion capacitance Smaller parasitic delay and lower dynamic power consumption Lower threshold voltages Select the correct answer using the code given below. 1, 2 and 3 1 and 2 only 1 and 3 only 2 and 3 only

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127. [UNKNOWN] unknown

122. The finite state machine in which the output is a function of the current state and inputs the output is a function of only the current state Which of the following machines are respectively correct for these styles? Mealy machine and Moore machine Moore machine and Mealy machine State machine and Mealy machine State machine and State machine

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128. [UNKNOWN] unknown

123. In EPROMs, applying a high voltage to the upper gate causes electrons to jump through the thin oxide onto the floating gate through the process known as mask programming one-time programming avalanche injection or Fowler-Nordheim tunneling erasing VBY-S-EDC/51A 22 ===== PAGE 23 =====

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129. [UNKNOWN] unknown

124. What is the range of values of a and b for which the linear time-invariant system with impulse response $n^0 n < 0$ is stable? Both $|a|$ and $|b| > 1$ are satisfied Both $|a| < 1$ and $|b| < 1$ are satisfied Both $|a| > 1$ and $|b| > 1$ are satisfied Both $|a| < 1$ and $|b| < 1$ are satisfied

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130. [UNKNOWN] unknown

125. The special case of a finite-duration A sequence is given as $x(n) = (2, 4, 0, 31$ The sequence $x(n)$ into a sum of weighted impulse sequences will be (a) $25(n+1) + 45(n) + 38(n-2)$ $25(n) + 46(n-1) + 38(n-3)$ $28(n) + 48(n-1) + 35(n-2)$ $23(n+1) + 45(n) + 36(n-1)$

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131. [UNKNOWN] unknown

126. The two advantages of FIR filters over IIR filters are they are guaranteed to be stable and non-linear they are marginally stable and linear they are guaranteed to be stable and may be constrained to have linear phase they are marginally stable and non-linear

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132. [UNKNOWN] unknown

127. The frequency response and the main lobe width for rectangular window are $N \cdot w(a) (6) \sin^2$ and $4n$ —It $2rc \ 81z \ 2 \ .0) \ N \sin^2 \ 03 \ 2 \sin- \ N \ . \ co \ sin^2 \ . \ oN \ sin - \ 4 \ 2$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRONICS-TELECOMMUNICATION-ENGINEERING-ESEP-19_0.pdf

133. [UNKNOWN] unknown

128. A controller that takes control of the buses and transfers data directly between source and destination bypassing the microprocessor is known as DMA controller read-write controller high-speed controller master-slave controller

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134. [UNKNOWN] unknown

129. A 2-byte instruction which accepts the data from the input port specified in the second byte and loads into the accumulator is OUT <8-bit port address> IN <8-bit port address> OUT R <8-bit port address> IN R <8-bit port address>
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135. [UNKNOWN] unknown

130. Consider the following instruction : EI MV! A, 08H SIM It means disable all interrupts enable all interrupts disable RST 7.5 and 6.5 enable RST 7.5 and 6-5

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRONICS-TELECOMMUNICATION-ENGINEERING-ESEP-19_0.pdf

136. [UNKNOWN] unknown

131. The instruction BC 0x15 means jump 15 bytes relative to the program counter copy and load 15 words in reverse direction to the program counter move to a location by 15 bits to the program counter redirect (jump) to a location by 15 words relative to the program counter

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRONICS-TELECOMMUNICATION-ENGINEERING-ESEP-19_0.pdf

137. [UNKNOWN] unknown

132. Which of the following constraints are to be considered by the designer while designing an embedded system?
Selecting the microcontroller as a controlling device
Selecting the language to write the software
Partitioning the tasks between hardware and software to optimize the cost

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRONICS-TELECOMMUNICATION-ENGINEERING-ESEP-19_0.pdf

138. [UNKNOWN] unknown

133. Which one of the following is the correct combination for a layer providing a service by means of primitives in an open systems interconnection? Request, Indication, Response and Confirm Request, Inform, Response and Service Request, Command, Response and Action Request, Confirm, Indication and Action

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRONICS-TELECOMMUNICATION-ENGINEERING-ESEP-19_0.pdf

139. [UNKNOWN] unknown

134. A network uses a fully interconnected mesh topology to connect 10 nodes together. The number of links required will be 35 40 45 50

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140. [UNKNOWN] unknown

135. Which of the following are the advantages of packet switching? Greater link efficiency than circuit switching
Connections are not blocked when traffic congestion occurs
Direct channel established between transmitter and receiver
No time is taken to establish connection
Select the correct answer using the code given below. Select the correct answer using the code given below. (a) 1,2 and 3 (a) 1 and 3 (b) 1 and 2 only (b) 1 and 2 (c) 1 and 3 only (c) 2 and 3 (d) 2 and 3 only (d) 3 and 4
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141. [UNKNOWN] unknown

136. A message consisting of 2400 bits is to

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142. [UNKNOWN] unknown

138. Which of the following statements are be passed over an internet. The message correct regarding CDMA? is passed to the transport layer which

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143. [UNKNOWN] unknown

139. Which of the following regarding cellular systems with small cells are correct?

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144. [UNKNOWN] unknown

140. A satellite is orbiting in the equatorial

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145. [UNKNOWN] unknown

141. A single-mode optical fiber has a beat length of 8 cm at 1300 nm. The value of birefringence B_f will be nearly

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146. [UNKNOWN] unknown

142. Which one of the following instruments is useful while measuring the optical power as a function of wavelength? (a) 6.3 km (a) Optical power attenuator (b) 5.2 km Optical power meter 4.1 km (c) Optical spectrum analyzer (d) Optical return loss tester (d) 3.0 km VBY-S-ELX/51A 26 ===== PAGE 27 ===== Directions : The following six (6) items consist of two statements, one labelled as 'Statement (I)' and the other as 'Statement (W). You are to examine these two statements carefully and select the answers to these items using the code given below : Code: Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I) Both Statement (I) and Statement (II) are individually true but Statement (II) is not the correct explanation of Statement (I) Statement (I) is true but Statement (II) is false Statement (I) is false but Statement (H) is true Statement (I) : Sign-magnitude representation is rarely used in implementing the integer portion of the ALU. Statement (II) : There are two representations of zero in sign-magnitude representation. Statement (I) : Dynamic loading gives better memory- space utilization. Statement (II) : In dynamic loading, an unused routine is never loaded. Statement (I) : SRAM is used for cache memory and DRAM is used for main memory. Statement (II) : SRAM is somewhat faster than DRAM.

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147. [UNKNOWN] unknown

143. The optical performance monitoring plane with a period from perigee to involves perigee of 12 h. If the eccentricity $= 0.002$, $i = 0^\circ$, $= 66063.17 \text{ km}^2$, $= 3.99 \times 10^{14} \text{ m}^3 / 82$ and the earth's (a) transport layer monitoring, optical signal monitoring and protocol equatorial radius $= 6378.14 \text{ km}$, the performance monitoring semi-major axis will be (a) 34232 km (13) 30424 km 26612 km 22804 km

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRONICS-TELECOMMUNICATION-ENGINEERING-ESEP-19_0.pdf

148. [UNKNOWN] unknown

144. An earth station at sea level communicates at an elevation angle of 35° with GEO satellite. The vertical height of the stratiform rain is 3 km. The physical path length L through the rain will be nearly

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRONICS-TELECOMMUNICATION-ENGINEERING-ESEP-19_0.pdf

149. [UNKNOWN] unknown

148. Statement (I) : In a multiuser system, each user is assigned a section of usable memory area and is not allowed to go out of the assigned memory area. Statement (II) : In multiuser system, there is a software mechanism to prevent unauthorized access of memory by different users. Statement (I) : The external surface of a crystal is an imperfection in

itself as the atomic bonds do not extend beyond the surface. Statement (II) : The external surfaces have surface energies that are related to the number of bonds broken at the surface. Statement (I) : By organizing various 'optical functions' into an 'array structure' via nano-pattern replication, 'spatial integration' is established. Statement (II) : By adding a nano-optic layer or layers to functional optical materials, the 'hybrid integration' is possible to be achieved. VBY-S-ELX/ 51A 27 [P.T.O.

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150. [UNKNOWN] unknown

170 - - — e RC E —e RC (d) (— + — go e RC R RC)

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